JUDE RUSSELL

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EDUCATION

Imperial College London Sept 2025 – Sept 2026
MSc in Physics with Quantum Dynamics

King's College London Sept 2021 – May 2025
BSc in Physics with a Year Abroad, First Class Honours

University of California, Berkeley Aug 2023 – May 2024
Physics exchange year

RESEARCH EXPERIENCE

Quantum Measurement Lab, Imperial College London Apr 2026 – Sept 2026
Master's Research Project – Dr Michael Vanner

- Measured the multimode Brillouin spectrum of standard optical fibre using heterodyne spectroscopy, resolving five closely spaced acoustic resonances.
- Developed analytical and numerical models of guided acoustic modes to investigate acoustic confinement and the physical origins of Brillouin resonances.
- Investigated the potential of multimode Brillouin spectroscopy for characterising acoustic properties of optical fibres. Further details of the research to be outlined in a forthcoming publication.

Atto@Kings, King's College London Jan 2025 – Apr 2025
Undergraduate Research Project – Dr Emilio Pisanty and Dr Anne Weber

- Extended a numerical simulation framework to investigate electron trajectories and quantum-path phase coefficients in two-colour high-harmonic generation.
- Calculated intensity-dependent phase responses and investigated possible signatures of quantum tunnelling without a classical barrier.
- Identified distinctive phase-sensitive features in bichromatic fields, providing possible directions for subsequent experimental investigation.

Levitated Nanophysics Group, King's College London Jun 2023 – Aug 2023
King's Undergraduate Research Fellowship – Professor James Millen

- Investigated the gradual loss of trapping stability of microscopic diamond particles in a modified Paul trap.
- Developed MATLAB particle-tracking software and used power spectral analysis to characterise the evolution of particle motion.
- Used simplified numerical modelling to investigate whether a reduction in charge-to-mass ratio could explain the observed frequency shift.

SELECTED PROJECTS

Laser spectroscopy for magneto-optical trapping Jul 2026
Experimental project, Imperial College London

- Aligned free-space optics and recorded Doppler-free saturated-absorption spectra of the rubidium D_2 line. Investigated laser-frequency drift in a system used for cooling-laser stabilisation.

Surface–GKP Quantum Error Correction Jan 2026 – Mar 2026
Technical review, Imperial College London

- Reviewed GKP encoding, syndrome extraction, decoding, finite squeezing and the concatenation of bosonic GKP codes with surface codes, including possible experimental implementations.

Ground-state energy estimation with SQD Oct 2025 – Nov 2025

IBM Qiskit Fall Fest Team Project

- Applied sample-based quantum diagonalisation to estimate the electronic ground-state energy of a water molecule.
- Compared the resulting estimates with a classical reference and presented the project at an Institute of Physics event.

Cal Aerospace SAE, UC Berkeley Sept 2023 – May 2024

Structures Team

- Contributed to the structural design and analysis of a 15 ft competition aircraft, using SolidWorks and Ansys finite-element analysis to investigate landing-gear performance under heavy landing loads.

In2scienceUK Jun 2020 – Aug 2020

Summer Intern

- Investigated the optical properties of lead-halide perovskites and analysed spectral data to identify band-gap trends.

PRESENTATIONS

Oral presentations

Surface-GKP Quantum Computing: Encoding a qubit in a oscillator Apr 2026

Imperial College London

Quantum computing for drug discovery Nov 2025

Institute of Physics, London

Quantum tunnelling without a barrier: phase-sensitive signatures in HHG May 2025

King's College London

Dancing levitated diamonds in a modified Paul trap Aug 2023

Levitated Nanophysics Group, King's College London

Poster presentations

Multimode Brillouin scattering in optical fibre Aug 2026

Blackett Laboratory Foyer, Imperial College London

Quantum tunnelling without a barrier? May 2025

Undergraduate poster conference, King's College London

TECHNICAL SKILLS

Programming: Python, Julia, MATLAB, Mathematica, LaTeX, HTML.

Experimental: Optical alignment, laser spectroscopy, heterodyne detection, photodetection, electrical spectrum analysis and high-voltage electronics.

Numerical methods: Eigenvalue calculations, finite-element modelling, Fourier analysis, optimisation, particle tracking and scientific data analysis.

Quantum software: Qiskit, QuTiP, IBM Quantum Platform.

AWARDS & FUNDING

King's Research Award Dec 2023

King's College London – undergraduate research

King's Undergraduate Research Fellowship May 2023

Competitive funded summer research programme

Turing Fund May 2023

Funding for study abroad in the United States

Quantum Mechanics Module Prize Dec 2022

King's College London – top performance in a cohort of 120